

LEVEL 3 — Pandas DataFrame Basics

20 Topics — Easy Line-by-Line Code, Output & Interview Explanations

1. What is DataFrame?

A DataFrame is a two-dimensional labeled table with rows and columns.

Code:

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["Amit", "Neha"],
    "Marks": [85, 90]
})

print(df)
```

Output:

	Name	Marks
0	Amit	85
1	Neha	90

Interview: DataFrame = table. It can contain multiple columns and different data types.

2. Creating DataFrame

Use `pd.DataFrame()` to create a DataFrame.

Code:

```
data = [[10, 20],
        [30, 40]]

df = pd.DataFrame(data)

print(df)
```

Output:

	0	1
0	10	20
1	30	40

Interview: Each inner list becomes one row.

3. DataFrame from Dictionary

Dictionary keys become column names and dictionary values become column data.

Code:

```
data = {
    "Name": ["Amit", "Neha", "Raj"],
    "Marks": [80, 90, 75]
}

df = pd.DataFrame(data)
print(df)
```

Output:

	Name	Marks
0	Amit	80
1	Neha	90
2	Raj	75

Interview: Key = column name. Value = column data.

4. DataFrame from List

A list of lists can be converted into a DataFrame.

Code:

```
data = [
    ["Amit", 80],
    ["Neha", 90],
    ["Raj", 75]
]

df = pd.DataFrame(data, columns=["Name", "Marks"])
print(df)
```

Output:

	Name	Marks
0	Amit	80
1	Neha	90
2	Raj	75

Interview: Each inner list represents one row.

5. DataFrame from NumPy Array

A 2D NumPy array can be converted into a DataFrame.

Code:

```
import numpy as np

arr = np.array([
    [10, 20],
    [30, 40],
    [50, 60]
])

df = pd.DataFrame(arr, columns=["A", "B"])
print(df)
```

Output:

	A	B
0	10	20
1	30	40
2	50	60

Interview: NumPy provides the array; Pandas adds row and column labels.

6. DataFrame from List of Dictionaries

Each dictionary represents one row. Keys become columns.

Code:

```
data = [
    {"Name": "Amit", "Marks": 80},
    {"Name": "Neha", "Marks": 90},
    {"Name": "Raj", "Marks": 75}
]

df = pd.DataFrame(data)
print(df)
```

Output:

	Name	Marks
0	Amit	80
1	Neha	90
2	Raj	75

Interview: This format is useful for record-by-record data.

7. Custom Index

You can give your own row labels using index=.

Code:

```
data = {
    "Name": ["Amit", "Neha"],
    "Marks": [80, 90]
}

df = pd.DataFrame(
    data,
    index=["Student1", "Student2"]
)
```

```
print(df)
```

Output:

	Name	Marks
Student1	Amit	80
Student2	Neha	90

Interview: Custom index replaces the default 0, 1, 2... labels.

8. Column Names

Use `columns=` to give meaningful names to columns.

Code:

```
data = [[10, 20], [30, 40]]

df = pd.DataFrame(
    data,
    columns=["Math", "Science"]
)

print(df)
```

Output:

	Math	Science
0	10	20
1	30	40

Interview: `columns=` assigns the column names.

9. index

The `index` property shows row labels.

Code:

```
df = pd.DataFrame({
    "Name": ["Amit", "Neha", "Raj"],
    "Marks": [80, 90, 75]
})

print(df.index)
```

Output:

```
RangeIndex(start=0, stop=3, step=1)
```

Interview: `index` tells how rows are labeled.

10. columns

The `columns` property shows all column names.

Code:

```
print(df.columns)
```

Output:

```
Index(['Name', 'Marks'], dtype='object')
```

Interview: `columns` tells which columns are present.

11. values

The `values` property returns the data without row and column labels.

Code:

```
print(df.values)
```

Output:

```
[['Amit' 80]
 ['Neha' 90]
 ['Raj' 75]]
```

Interview: `values` gives the underlying data representation.

12. shape

shape returns the number of rows and columns.

Code:

```
print(df.shape)
```

Output:

```
(3, 2)
```

Interview: (3, 2) means 3 rows and 2 columns.

13. size

size returns the total number of cells.

Code:

```
print(df.size)
```

Output:

```
6
```

Interview: $3 \times 2 = 6$ cells.

14. ndim

ndim returns the number of dimensions.

Code:

```
print(df.ndim)
```

Output:

```
2
```

Interview: A DataFrame is two-dimensional.

15. dtypes

dtypes shows the data type of each column.

Code:

```
print(df.dtypes)
```

Output:

```
Name      object
Marks     int64
dtype: object
```

Interview: Each column can have a different data type.

16. info()

info() gives a quick structural summary: rows, columns, non-null values, data types and memory information.

Code:

```
df.info()
```

Output:

```
Output example:
RangeIndex: 3 entries, 0 to 2
Data columns (total 2 columns):
Name      3 non-null object
Marks     3 non-null int64
dtypes: int64(1), object(1)
```

Interview: info() is very useful for understanding a dataset and finding missing values.

17. describe()

describe() gives statistical information for numeric columns.

Code:

```
print(df.describe())
```

Output:

	Marks
count	3.000000
mean	81.666667
std	7.637626
min	75.000000
25%	77.500000
50%	80.000000
75%	85.000000
max	90.000000

Interview: describe() gives count, mean, std, quartiles, min and max.

18. head()

head() displays the first 5 rows by default. You can specify a number.

Code:

```
print(df.head(2))
```

Output:

	Name	Marks
0	Amit	80
1	Neha	90

Interview: head(2) means first 2 rows.

19. tail()

tail() displays the last 5 rows by default.

Code:

```
print(df.tail(2))
```

Output:

	Name	Marks
1	Neha	90
2	Raj	75

Interview: tail(2) means last 2 rows.

20. sample()

sample() returns random rows from the DataFrame.

Code:

```
print(df.sample(2, random_state=1))
```

Output:

Example output:

	Name	Marks
0	Amit	80
2	Raj	75

Interview: sample(2) selects 2 random rows. random_state makes the result reproducible.

Quick Interview Revision

Q: What is a DataFrame?

A: A two-dimensional labeled data structure with rows and columns.

Q: How do you create a DataFrame?

A: pd.DataFrame(data).

Q: Dictionary to DataFrame?

A: Dictionary keys become columns and values become column data.

Q: What does shape return?

A: (rows, columns).

Q: What does size return?

A: Total number of cells.

Q: What does ndim return?

A: Number of dimensions; DataFrame = 2.

Q: What does dtypes show?

A: Data type of each column.

Q: What does info() do?

A: Shows structure, non-null counts, data types and memory information.

Q: What does describe() do?

A: Provides descriptive statistics for numeric columns.

Q: head() vs tail()?

A: head() shows the beginning; tail() shows the end.

Q: What does sample() do?

A: Returns random rows.

Final Practice Program

```
import pandas as pd

data = {
    "Name": ["Amit", "Neha", "Raj", "Priya"],
    "Age": [22, 21, 23, 22],
    "Marks": [80, 92, 75, 88]
}

df = pd.DataFrame(data)

print(df)
print("Shape:", df.shape)
print("Size:", df.size)
print("Dimensions:", df.ndim)
print("Columns:", df.columns)
print("Data types:")
print(df.dtypes)
print("First 2:")
print(df.head(2))
print("Last 2:")
print(df.tail(2))
print("Random 2:")
print(df.sample(2, random_state=1))
print("Statistics:")
print(df.describe())
```

Easy memory trick: DataFrame = TABLE. Properties: index, columns, values, shape, size, ndim, dtypes.
Inspection: info(), describe(), head(), tail(), sample().